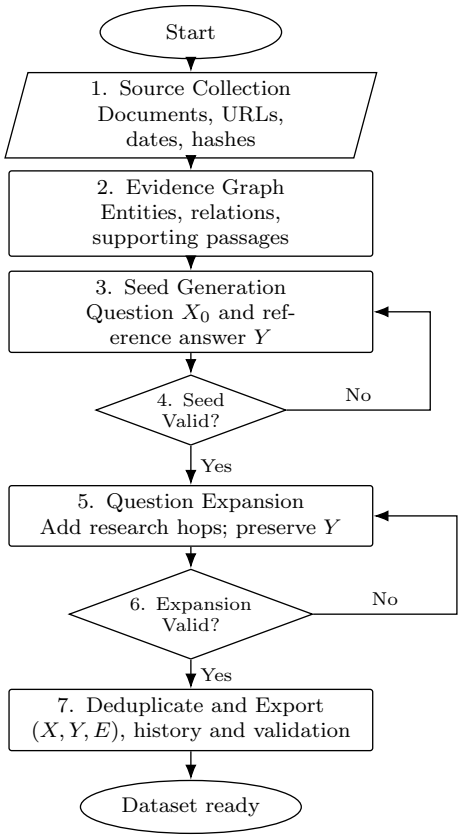


Figure 1: Synthetic deep-research data generation with independent validation and iterative refinement.